MUHAMMAD ZAIN-UL-ABIDEEN

AI Engineer

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Summary

AI Engineer with two ML/AI internships and deployed projects in speech recognition, computer vision, NLP, RAG, and LLM agents, plus an IEEE-format research paper under review. Builds systems end to end and evaluates them on held-out data.

Education

National University of Sciences and Technology (NUST) Sep 2022 – Jun 2026
Bachelor of Electrical Engineering Islamabad, Pakistan

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, Data Structures & Algorithms
Machine Learning: Machine Learning, Deep Learning, NLP, Computer Vision, PyTorch, Scikit-learn, Hugging Face Transformers, CLIP, Speech Recognition (Whisper), ONNX Runtime, Pandas, NumPy, Model Evaluation
LLMs & Retrieval: LLM Applications, RAG, AI Agents, Multi-Agent Systems, Prompt Engineering, Embeddings, Vector Search (pgvector, FAISS), Hybrid Search (BM25), Rerankers, LLM Evaluation, Gemini, Groq, Ollama, Langfuse
Backend & Cloud: FastAPI, REST APIs, PostgreSQL, SQLAlchemy, Next.js, Playwright, Docker, GitHub Actions (CI/CD), pytest, Git, Microsoft Azure, Google Cloud (GCP), Render, Vercel

Projects

Earshot – Podcast Q&A with Audio Citations | Python, Whisper, pgvector, PostgreSQL earshot-ten-pi.vercel.app
- Built a speech recognition and RAG system that transcribes podcasts with Whisper large-v3-turbo (2.65% WER) and links every answer to the exact second it was said in the original audio.
- Benchmarked keyword, vector, hybrid, and reranked search over 875 passages: a reranker lifted recall@5 from 0.76 to 0.93 (MRR 0.60 to 0.85) but cost 1.35 s vs. 69 ms latency; shipped hybrid and documented the trade-off.
- Added a verbatim quote checker with a repair pass that raised answered questions from 77.8% to 93.3% and correctly timed citations from 62.2% to 80.0%.

Dehqan – Crop Disease Detection for Pakistani Farmers | Python, PyTorch, CLIP, ONNX dehqan.vercel.app
- Trained a 27-condition, 13-crop leaf-disease deep learning classifier and exposed its lab-to-field gap (95.5% on lab photos, 11.5% in real fields); a CLIP embedding with a trained head raised held-out field accuracy to 61.6% in int8 ONNX (63.8% in PyTorch) vs. a 7.4% baseline.
- Shipped selective prediction that defers to extension offices below 0.9 confidence (answers 56% of field photos at 81% accuracy), a zero-shot CLIP leaf gate (99.8% of leaves accepted vs. 23.6% using classifier confidence), and an Urdu/English NLP question router (90% on a frozen test set) that refused 7/7 pesticide-dose requests.

EdgarIQ – SEC Filings Copilot | Python, SEC EDGAR API, Ollama, Groq, JavaScript edgar-iq-web.vercel.app
- Built a multi-agent RAG system over live SEC EDGAR filings (planner, retriever, drafter, critic) with a numeric grounding checker that verifies every generated figure against retrieved source text.
- Diagnosed wrong-quarter retrieval in pure semantic search and designed a hybrid metadata-filtered retriever that raised golden-set accuracy from 29% to 100%; built an LLM-as-judge eval harness backed by 89 unit/integration tests.

Rehnuma – AI Copilot for Electricity Bills | Python, FastAPI, Next.js, Gemini, Groq rehnuma-kappa.vercel.app
- Built an Urdu/English bill auditor where a deterministic engine re-checks every figure a Gemini vision model reads, triggering targeted re-reads (96.2% field accuracy on real photos; 0 invented numbers reached users).
- Developed hybrid RAG (BM25 + character n-grams, RRF) over 566 regulatory clauses with a citation critic; LLM query rewriting lifted Urdu top-5 retrieval from 0% to 83% (91% held-out English); 319 tests and 20 eval metrics gated in CI.

Experience

Machine Learning Engineer Intern Jun 2025 – Aug 2025
RISETech Islamabad, Pakistan
- Led feature extraction for a multi-sensor (vibration, magnetic, acoustic) vehicle classifier; trained ensemble machine learning models (Random Forest, SVM, KNN, LDA, Extra Trees) in Scikit-learn that raised accuracy from 71% to 99%.
- Authored an IEEE-format research paper on the feature extraction method (under review).

AI Engineer Intern Jun 2024 – Aug 2024
Software Productivity Strategists (SPS), NSTP Islamabad, Pakistan
- Developed Python backend features for two flagship products (BMS, CSM) and deployed AI solutions on Microsoft Azure.

Certifications

- Google AI Professional Certificate • Advanced ML on Google Cloud Specialization • IBM Machine Learning Professional Certificate • AWS Cloud Practitioner Essentials